



Qualification Pack



Locomotive Driver

QP Code: ISC/Q0402

Version: 1.0

NSQF Level: 4

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ISC/Q0402: Locomotive Driver

Brief Job Description

The job is all about operating the locomotive to ensure movement of raw materials, intermediates, finished products within the plant. It includes maintaining log book for performance parameters and working condition of locomotive.

Personal Attributes

The job requires the candidate to work independently as well as in teams. Some of the key attributes includes physical fitness, not having colour blindness, having analytical skills, problem solving attitude, disciplined, high concentration levels, a sharp reflex and willingness to work in a factory environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0009: Work effectively with others](#)
3. [ISC/N0404: Inspect critical systems of locomotive](#)
4. [ISC/N0405: Operate locomotive to carry material](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Iron Making
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL



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Minimum Educational Qualification & Experience	10th Class with 0-6 Months of experience & In lieu of minimum qualification the incumbent should not be considered for this role & Driving experience and minimum 100 hours hands on experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	One month training in the Loco shop Routine Inspection of locomotive Understanding of traffic signals 5S and safety practices
Minimum Job Entry Age	18 Years
Last Reviewed On	30/12/2014
Next Review Date	31/03/2022
NSQC Approval Date	20/07/2015
Version	1.0
Reference code on NQR	2015/IS/IISSSC/00767
NQR Version	1.0



Qualification Pack

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Use protective clothing/equipment for specific tasks and work conditions
Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks
Glasses/gloves/visors
Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator

Health and safety procedures

To be competent, the user/individual on the job must be able to:

- PC2.** State the name and location of people responsible for health and safety in the workplace
Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.
- PC3.** State the names and location of documents that refer to health and safety in the workplace
- PC4.** Identify job-site hazardous work and state possible causes of risk or accident in the workplace
Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.
- PC5.** Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role
Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors
- PC6.** State location of general health and safety equipment in the workplace



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- PC7.** Inspect for faults, set up and safely use steps and ladders in general use
Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc.
Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.
- PC8.** Work safely in and around trenches, elevated places and confined areas
- PC9.** Lift heavy objects safely using correct procedures
- PC10.** Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces
- PC11.** Identify common hazard signs displayed in various areas
- PC12.** Retrieve and/or point out documents that refer to health and safety in the workplace

Fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC13.** Use the various appropriate fire extinguishers on different types of fires correctly.
Fire extinguishers: Sand Water Foam Co2 Dry powder
Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents)
Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.
- PC14.** Demonstrate rescue techniques applied during fire hazard
- PC15.** Demonstrate good housekeeping in order to prevent fire hazards
- PC16.** Demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC17.** Demonstrate how to free a person from electrocution
- PC18.** Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC19.** Demonstrate basic techniques of bandaging
- PC20.** Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below :
Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.
- PC21.** Perform and organize loss minimization or rescue activity during an accident in real or simulated environments
- PC22.** Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC23.** Demonstrate the artificial respiration and the CPR Process
- PC24.** Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work



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- PC25.** Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer
- PC26.** Demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace
- KU2.** State the names and location of documents that refer to health and safety in the workplace
- KU3.** Meaning of hazards and risks
- KU4.** Health and safety hazards commonly present in the work environment and related precautions
- KU5.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU6.** Activities and causes of risk and accident
- KU7.** Methods of accident prevention
- KU8.** Safe working practices when working with tools and machines
- KU9.** Safe working practices while working at various hazardous sites
- KU10.** Where to find all the general health and safety equipment in the workplace
- KU11.** Various dangers associated with the use of electrical equipment
- KU12.** Preventative and remedial actions to be taken in the case of exposure to toxic materials. Exposure: ingested, contact with skin, inhaled Preventative action: ventilation, masks, protective clothing/equipment Remedial action: immediate first aid, report to supervisor Materials: solvents, flux, lead
- KU13.** Importance of using protective clothing/equipment while working
- KU14.** Precautionary activities to prevent the fire accident Activities and causes: Physical actions Reading Listening to and giving instructions Inattention Sickness and incapacity (e.g. Drunkenness) Health hazards (e.g. Untreated injuries and contagious illness)
- KU15.** Various causes of fire
- KU16.** Techniques of using the different fire extinguishers
- KU17.** Different methods of extinguishing fire
- KU18.** Rescue techniques applied during a fire hazard
- KU19.** Various types of safety signs and what they mean
- KU20.** Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU21.** Content of written accident report



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- KU22.** Potential injuries and ill health associated with incorrect manual handling
- KU23.** Safe lifting and carrying practices
- KU24.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU25.** Potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and comprehend basic content to read labels, charts, signages
- GS2.** Read and comprehend basic English to read manuals of operations
- GS3.** Read and write an accident/incident report in local language or English
- GS4.** Question co-workers appropriately in order to clarify instructions and other issues
- GS5.** Give clear instructions to co-workers, subordinates others
- GS6.** Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
- GS7.** Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** Remain congenial while discussing and debating issues with co-workers
- GS9.** Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS10.** Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
- GS11.** Thank co-workers for any assistance received
- GS12.** Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
- GS13.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS14.** Identify immediate or temporary solutions to resolve delays
- GS15.** Identify sources of support that can be availed of for problem solving for various kind of problems
- GS16.** Report problems that you cannot resolve to appropriate authority
- GS17.** Identify cause and effect relations in their area of work
- GS18.** Use cause and effect relations to anticipate potential problems and their solution



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	5	5	-	-
PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks Glasses/gloves/visors Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator	5	5	-	-
<i>Health and safety procedures</i>	15	40	-	-
PC2. State the name and location of people responsible for health and safety in the workplace Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.	-	4	-	-
PC3. State the names and location of documents that refer to health and safety in the workplace	-	1	-	-
PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors	5	5	-	-
PC6. State location of general health and safety equipment in the workplace	-	5	-	-
PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc. Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.	-	5	-	-
PC8. Work safely in and around trenches, elevated places and confined areas	-	4	-	-
PC9. Lift heavy objects safely using correct procedures	-	4	-	-
PC10. Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces	-	1	-	-
PC11. Identify common hazard signs displayed in various areas	5	1	-	-
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	-	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fire safety procedures</i>	10	15	-	-
PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: Sand Water Foam Co2 Dry powder Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.	5	5	-	-
PC14. Demonstrate rescue techniques applied during fire hazard	5	5	-	-
PC15. Demonstrate good housekeeping in order to prevent fire hazards	-	1	-	-
PC16. Demonstrate the correct use of a fire extinguisher	-	4	-	-
<i>Emergencies, rescue and first-aid procedures</i>	15	45	-	-
PC17. Demonstrate how to free a person from electrocution	-	5	-	-
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	5	5	-	-
PC19. Demonstrate basic techniques of bandaging	-	5	-	-
PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments	-	5	-	-
PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	-	5	-	-
PC23. Demonstrate the artificial respiration and the CPR Process	-	5	-	-
PC24. Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work	-	4	-	-
PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer	5	5	-	-
PC26. Demonstrate correct method to move injured people and others during an emergency	-	1	-	-
NOS Total	45	105	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0009: Work effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace

Elements and Performance Criteria

Ensure appropriate communication with superiors, peers and others as applicable at work place

To be competent, the user/individual on the job must be able to:

- PC1.** Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** Provide information to others clearly, at a pace and in a manner that helps them to understand

Demonstrate appropriate behaviour and etiquette at work place

To be competent, the user/individual on the job must be able to:

- PC4.** Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** Display appropriate communication etiquette while working
- PC7.** Display active listening skills while interacting with others at work
- PC8.** Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** Demonstrate responsible and disciplined behaviours at the workplace
- PC10.** Escalate grievances and problems to

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** Reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** Relevant people and their responsibilities within the work area
- KU4.** Escalation matrix and procedures for reporting work and employment related issues
- KU5.** Various categories of people that one is required to communicate and co ordinate with in the organization
- KU6.** Importance of effective communication in the workplace
- KU7.** Importance of teamwork in organizational and individual success
- KU8.** Various components of effective communication



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- KU9.** Key elements of active listening
- KU10.** Value and importance of active listening and assertive communication
- KU11.** Barriers to effective communication
- KU12.** Importance of tone and pitch in effective communication
- KU13.** Importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** How poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** Importance of ethics for professional success
- KU16.** Importance of discipline for professional success
- KU17.** What constitutes disciplined behaviour for a working professional
- KU18.** Common reasons for interpersonal conflict
- KU19.** Importance of developing effective working relationships for professional success
- KU20.** Expressing and addressing grievances appropriately and effectively
- KU21.** Importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and comprehend basic content to read labels, charts, signages
- GS2.** Read and comprehend basic English to read manuals of operations
- GS3.** Read and write an accident/incident report in local language or English
- GS4.** Question co-workers appropriately in order to clarify instructions and other issues
- GS5.** Provide clear instructions to co-workers, subordinates others
- GS6.** Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down
- GS7.** Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** Remain congenial while discussing and debating issues with co-workers
- GS9.** Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS10.** Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
- GS11.** Thank co-workers for any assistance received
- GS12.** Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
- GS13.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS14.** Identify immediate or temporary solutions to resolve delays
- GS15.** Identify sources of support that can be availed of for problem solving for various kind of problems



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- GS16.** Identify sources of support that can be availed of for problem solving for various kind of problems
- GS17.** Report problems that you cannot resolve to appropriate authority
- GS18.** Identify cause and effect relations in their area of work
- GS19.** Use cause and effect relations to anticipate potential problems and their solution



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure appropriate communication with superiors, peers and others as applicable at work place</i>	10	20	-	-
PC1. Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	5	5	-	-
PC2. Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	5	5	-	-
PC3. Provide information to others clearly, at a pace and in a manner that helps them to understand	-	10	-	-
<i>Demonstrate appropriate behaviour and etiquette at work place</i>	20	50	-	-
PC4. Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible	5	5	-	-
PC5. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	5	5	-	-
PC6. Display appropriate communication etiquette while working	-	10	-	-
PC7. Display active listening skills while interacting with others at work	-	10	-	-
PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	5	5	-	-
PC9. Demonstrate responsible and disciplined behaviours at the workplace	5	10	-	-
PC10. Escalate grievances and problems to	-	5	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0009
NOS Name	Work effectively with others
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0404: Inspect critical systems of locomotive

Description

This unit is about inspecting the locomotive to ensure they are in working condition before operating it for transfer of material

Elements and Performance Criteria

Take handover from previous shift operator

To be competent, the user/individual on the job must be able to:

- PC1.** report on the job as per schedule given by shift in charge
- PC2.** take over from the previous shift operator and understand the task to be done safely
- PC3.** zero derailment /collision and no accident

Conduct physical inspections to check working condition of locomotive

To be competent, the user/individual on the job must be able to:

- PC4.** conducts physical inspections to check coolant level, oil level etc. as per checklist
- PC5.** report deviation from desirable levels including rail track , if any to shift in charge to get them rectified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality and damage checks to be done and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** escalation matrix for reporting identified issues
- KU4.** knowledge of area of operation
- KU5.** knowledge of associated equipment, accessories and components
- KU6.** checks that need to be made to ensure that equipment is safe and ready to use
- KU7.** knowledge of traffic signals and signs
- KU8.** understanding of normal running characteristics of locomotive
- KU9.** possible causes of common problems during operation of locomotive and remedies
- KU10.** implications of not adhering to sequence of activities and operations
- KU11.** response to emergencies e.g. power failures ,fire and system failures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** express statements, opinions or information clearly so that others can hear and understand
- GS2.** respond appropriately to signals
- GS3.** communicate with supervisor



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- GS4.** communicate with team members
- GS5.** handle different machines and tools associated with locomotive
- GS6.** knowledge of parts of locomotive and their functioning
- GS7.** diagnose common problems in the tools based on visual inspection, sound, temperature etc
- GS8.** detect emergencies and rectify the same



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Take handover from previous shift operator</i>	30	75	-	-
PC1. report on the job as per schedule given by shift in charge	10	25	-	-
PC2. take over from the previous shift operator and understand the task to be done safely	10	25	-	-
PC3. zero derailment /collision and no accident	10	25	-	-
<i>Conduct physical inspections to check working condition of locomotive</i>	20	175	-	-
PC4. conducts physical inspections to check coolant level, oil level etc. as per checklist	10	100	-	-
PC5. report deviation from desirable levels including rail track , if any to shift in charge to get them rectified	10	75	-	-
NOS Total	50	250	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0404
NOS Name	Inspect critical systems of locomotive
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Iron Making
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015



Qualification Pack

ISC/N0405: Operate locomotive to carry material

Description

This unit is about operating the locomotive as per the Standard Operating Procedure (SOP) carry material within the plant

Scope

This unit/task covers the following:

- Operate locomotive to transfer material
- Record keeping according to specifications

Elements and Performance Criteria

Operate locomotive to transfer material

To be competent, the user/individual on the job must be able to:

- PC1.** be aware of how much material can be safely carried according to the locomotive's capacity
- PC2.** start the engine of the locomotive as per standard operating procedure (sop)
- PC3.** check the parameters from the panel oil pressure, air pressure, charging system, brakes etc.
- PC4.** adhere to traffic signals as given by the traffic department

Record keeping according to specifications

To be competent, the user/individual on the job must be able to:

- PC5.** keep track of critical indicators as specified in the sop
- PC6.** records all parameters according to specifications in the log book for the equipment
- PC7.** take loco to fuel station as per requirement
- PC8.** identify any deviations from specifications and report to shift in charge

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality and damage checks to be done and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** escalation matrix for reporting identified issues
- KU4.** types of documentation in organization and importance of the same
- KU5.** records to be maintained and implications of non-maintenance of the same
- KU6.** knowledge of area of operation and carrying material
- KU7.** knowledge of associated equipment, accessories and components
- KU8.** checks that need to be made to ensure that equipment is safe and ready to use
- KU9.** knowledge of traffic signals and signs
- KU10.** understanding of normal running characteristics of locomotive



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- KU11.** possible causes of common problems during operation of locomotive and remedies
- KU12.** implications of not adhering to sequence of activities and operations
- KU13.** response to emergencies e.g. power failures ,fire and system failures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** express statements, opinions or information clearly so that others can hear and understand
- GS2.** respond appropriately to signals
- GS3.** communicate with supervisor
- GS4.** communicate with team members
- GS5.** take responsibility for completing ones own work assignment
- GS6.** take initiative to enhance/learn skills in other areas of work
- GS7.** the capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse ones learning
- GS8.** is open to new ways of doing things
- GS9.** the capacity to envisage and articulate personal goals; to develop strategies and take action to achieve them
- GS10.** construct simple sentences and express ideas clearly through written communication
- GS11.** fill up appropriate technical forms, record keeping documents, activity logs in required format of the company
- GS12.** handle different machines and tools associated with locomotive
- GS13.** knowledge of parts of locomotive and their functioning
- GS14.** avoid absenteeism and maintain quality of work
- GS15.** work in a disciplined environment and adhere to working norms of the organisation
- GS16.** be punctual and adhere to timelines
- GS17.** diagnose common problems in the tools based on visual inspection, sound, temperature etc
- GS18.** detect emergencies and rectify the same



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate locomotive to transfer material</i>	40	310	-	-
PC1. be aware of how much material can be safely carried according to the locomotive's capacity	10	80	-	-
PC2. start the engine of the locomotive as per standard operating procedure (sop)	10	80	-	-
PC3. check the parameters from the panel oil pressure, air pressure, charging system, brakes etc.	10	80	-	-
PC4. adhere to traffic signals as given by the traffic department	10	70	-	-
<i>Record keeping according to specifications</i>	40	60	-	-
PC5. keep track of critical indicators as specified in the sop	10	15	-	-
PC6. records all parameters according to specifications in the log book for the equipment	15	15	-	-
PC7. take loco to fuel station as per requirement	5	10	-	-
PC8. identify any deviations from specifications and report to shift in charge	10	20	-	-
NOS Total	80	370	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0405
NOS Name	Operate locomotive to carry material
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Iron Making
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



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Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	45	105	-	-	150	15
ISC/N0009.Work effectively with others	30	70	-	-	100	10
ISC/N0404.Inspect critical systems of locomotive	50	250	-	-	300	30
ISC/N0405 Operate locomotive to carry material	80	370	-	-	450	45
Total	205	795	-	-	1000	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
5S	Technique of maintaining orderliness “Japanese terminology.
CP	Control Plan.
WI	Work Instructions.



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.